

FAECO: 面向预布局逻辑重综合的失效驱动候选生成引擎

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摘要: 工程变更命令修复中, 缓冲器插入与门尺寸调整仅能微调门级延迟, 对逻辑级数过深的关键路径无能为力; 逻辑重综合虽能改变电路结构, 但其候选构造阶段将等价性验证与物理时序优化串行处理, 导致验证失败与收益不足两类问题相互制约。

关键词: 时序 ECO; 重综合; 双目标割; 失效驱动反馈; 简化 SPEF 门控; 预布局逻辑重综合 中图法分类号: TP391.41

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FAECO: Failure-Aware ECO Candidate Generation Engine for Pre-Placement Logic Re-Synthesis

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Abstract: In engineering change order (ECO) repair, buffer insertion and gate sizing can only tune gate-level delays and cannot handle critical paths with excessive logic depth; logic re-synthesis can change the circuit structure, but its candidate construction stage processes equivalence checking and physical timing optimization serially, causing the mutual restriction of verification failures and insufficient timing gains. To address this, FAECO (Failure-Aware ECO), a failure-aware candidate generation engine for pre-placement logic re-synthesis, is proposed. It encodes both equivalence and timing preferences into cut weights, adaptively updates the weights and re-searches the cut according to six failure modes, and embeds estimated wire load into the inner-loop acceptance criterion to filter physically robust candidates via two-stage gating (an ideal-net improvement threshold and a simplified SPEF recheck). Strict timing improvements are achieved on ISCAS89 (8/8), ITC-99 (18/19), and PicoRV32 (2/3) benchmarks. Under the convergence configuration, the average WNS improvement is 1.07 ns (mean of per-circuit improvements), $6.7\times$ that of the gate-sizing-only method; under the efficiency-oriented configuration, only 27 STA calls are needed to complete the 8/8 repair, cutting the overhead by 99.1%. After real placement and routing, 5 of the 8 circuits retain the improvement, 1 ties, and 2 regress by no more than 0.02 ns. FAECO implements a closed-loop failure-driven logic re-synthesis repair on the open-source toolchain.

Key words: timing ECO; re-synthesis; bi-objective cut; failure-aware feedback; simplified SPEF gating; pre-placement logic re-synthesis

Body text.